

Initial m68k support for Zephyr

MC68000 + MC68010 · QEMU-first · working implementation

Embedded engineer in Finland · 22 years in embedded · retrocomputing

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Why this exists

Started as a pet project, gained upstream value

I wanted to put Zephyr to my CV. Combine this with retrocomputing, and I started making the port. So, it started as a personal m68k project.

It now gives Zephyr a reproducible target with a rare combination of architectural constraints.

The upstream case is portability — not nostalgia.

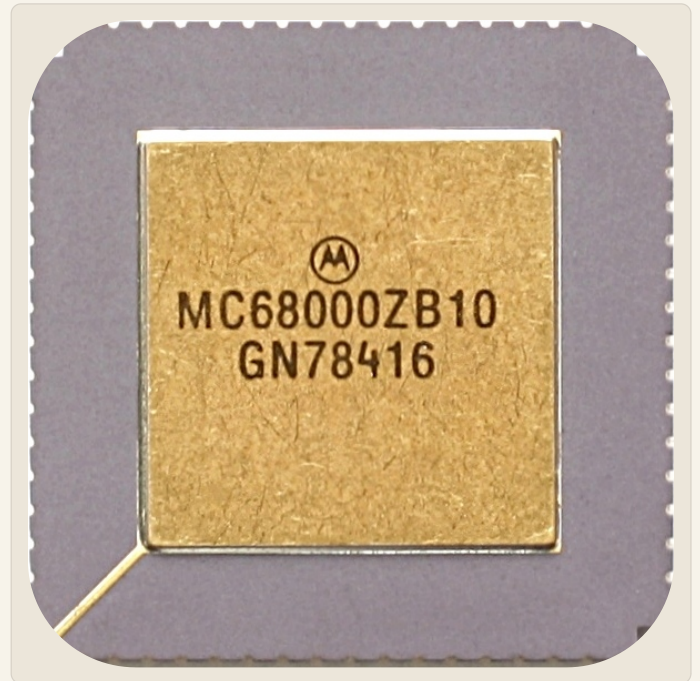


Photo: Konstantin Lanzet · CC BY 3.0

This is a working port

Working, not compile-only

QEMU virt boots and runs threads, interrupts and exceptions, timer, TLS, logging, shell, MCUmgr, and C++ on MC68000 and MC68010.

953

test suites executed

8,006

test cases passed

0

failed / errored / retried

Why m68k is useful to Zephyr

01

Big-endian

Tests byte-order assumptions that little-endian CI can hide.

03

Exception-frame alignment

Some frames are not multiples of four; a 68000 interrupt may leave SP only word-aligned.

02

Two-byte stack alignment

Stresses shared code that quietly assumes four-byte alignment.

04

A portability oracle

The port has already exposed generic kernel, IRQ, logging, and test assumptions.

Keep the initial scope deliberately narrow

In scope

MC68000 and MC68010

QEMU virt reference target

Uniprocessor; no MMU or FPU

Goldfish IRQ, RTC, and TTY

Runnable tests, docs, and ownership

Not in this RFC

68020+ and ColdFire

Physical boards

Networking, storage, and broad drivers

SMP and MMU-based userspace

FPU support

A mergeable integration path

The dependency chain and review boundaries are explicit.

01 · PREREQUISITES

Land shared enablers

SDK toolchain
Multilevel IRQ fix
sflist alignment

02 · CORE SERIES

Review in clear layers

Generic fixes → arch core
Goldfish drivers → QEMU
SoC/board

03 · QUALITY GATE

Agree the support bar

Tests and docs
Maintainers
Minimum CI subset